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Project Report

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Project Name:

*Photometric Classification of Supernova Subtypes from ZTF Light Curves: A
LightGBM–GRU Comparison with LSST-Compatible Preprocessing*

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Photometric Classification of Supernova Subtypes from ZTF Light Curves: A LightGBM–GRU Comparison with LSST-Compatible Preprocessing

Abstract

Time-domain surveys such as the Zwicky Transient Facility (ZTF) and the newly operational Vera C. Rubin Observatory (LSST) generate far more transient alerts than can be spectroscopically followed up, making fast, reliable photometric pre-classification a genuine operational bottleneck for the alert-broker ecosystem (Fink, ALeRCE, Lasair). This project builds and evaluates a compact, reproducible three-class (Type Ia, Ib/c, II) supernova classification pipeline on ZTF light curves, comparing a feature-engineered LightGBM baseline against a GRU sequence model, using a synthetic dataset for pipeline validation and multiple real, ALeRCE-sourced ZTF samples for the main study.

On a curated, class-balanced real sample (450 objects), LightGBM reached a macro-F1 of 0.985 against the GRU’s 0.956, both far above the 0.163 majority-class floor. Because this single-split result could reflect a favourable split, the comparison was repeated across 8 random seeds (LightGBM 0.945 ± 0.026 vs. GRU 0.843 ± 0.030) and on a naturally imbalanced sample (Ia:250/II:106/Ibc:94) and a $3\times$ -larger balanced sample (1,050 objects), all of which confirm LightGBM as the more robust performer on real ZTF data — a reversal of the ranking observed on synthetic data, where the GRU was stronger (0.878 vs. 0.832). An early-epoch ablation shows classification quality is overwhelmingly governed by how much of the light curve has been observed: macro-F1 roughly doubles from 0.65 at a 3-day cutoff to 0.99 at full length. A confidence/anomaly-flagging prototype was independently validated: prediction entropy correlates with correctness (ROC-AUC = 0.85 for predicting misclassification) and correctly flags 94% of injected synthetic out-of-distribution objects versus 10% of normal test objects. The results are read as a rigorous proof-of-concept for ZTF-based, LSST-schema-compatible photometric SN typing, not as an operational or spectroscopically validated classification system.

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1 Introduction, Literature Survey, Methodology, Motive/Objectives

1.1 Introduction

Supernovae (SNe) are broadly divided by their spectra into Type Ia (thermonuclear explosions of white dwarfs) and core-collapse types, of which Type Ib/c and Type II are the most common photometrically distinguishable classes. Because spectroscopic confirmation is slow and telescope time is scarce, wide-field time-domain surveys increasingly rely on photometric light curves — brightness measured over time in one or more filters — to pre-classify transients and prioritise which ones deserve costly spectroscopic follow-up. ZTF has produced this kind of alert stream since 2018, and the Rubin Observatory’s LSST, which issued its first public alerts in February 2026 and formally began its 10-year survey on 30 June 2026, is poised to increase the alert rate by roughly three orders of magnitude. This makes automated, photometric-only SN subtype classification an immediately relevant, currently active problem in observational astrophysics.

This project designs, implements, and rigorously evaluates such a classifier: a three-class (Ia / Ib–c / II) pipeline that compares a hand-engineered-feature gradient-boosted-tree model (LightGBM) against a learned sequence model (a GRU recurrent network) on real ZTF light curves obtained through the ALerCE broker, with a synthetic dataset used to validate the pipeline end-to-end before committing to real, noisier data.

1.2 Literature Survey

The proposed pipeline sits within an active ecosystem of astronomical alert brokers and classification challenges:

- **ALerCE** (Automatic Learning for the Rapid Classification of Events) is a Chilean-led broker that has processed the public ZTF alert stream since 2019, using a stamp-based classifier for rapid triage and a light-curve-based classifier for refined typing [4, 10, 2]. This project’s real data was obtained via ALerCE’s public query interface.
- **Fink** is an official LSST community broker that has processed the ZTF stream since November 2019 through independently-maintained “science modules”, including an early-Type-Ia classifier and, more recently, anomaly-detection modules [9]. Fink’s early-SN-Ia module has reported hundreds of candidates to the Transient Name Server, illustrating the real-world stakes of getting photometric pre-classification right.
- **Lasair** [11] is a further example of the broker paradigm this project’s preprocessing schema is designed to be conceptually compatible with.
- **PLAsTiCC / ELAsTiCC** [12, 8, 5] are public, simulated LSST-like light-curve datasets built specifically to benchmark photometric time-series classifiers ahead of real LSST data, and represent the standard fallback/benchmark dataset in this sub-field.
- **ZTF** itself [1] and the **Rubin Observatory/LSST** [6] are the underlying survey facilities whose alert schema (time, filter/band, flux, per-object detection sequence) this project’s data format deliberately mirrors.

Across this literature, two recurring modeling families dominate photometric SN classification: engineered-feature classifiers (random forests, gradient-boosted trees) operating on physically-motivated summary statistics, and sequence/deep-learning models operating more directly on the raw or lightly-processed flux time series. This project’s core methodological choice — comparing exactly these two families head-to-head on the same data and splits, rather than committing to one — follows directly from the fact that the literature has not settled which family is preferable, especially under the sparse, irregularly-sampled conditions typical of early-epoch or single-survey data.

Note on attribution. The survey above (broker descriptions, dataset provenance, and facility characteristics) reports facts and interpretations drawn from the cited external publications and is not original analysis. All data acquisition (via the ALeRCE public query interface), preprocessing, feature engineering, model training, ablations, statistical tests, and the discussion in Sections 3–5 constitute original work carried out for this project, and any external code or algorithm used (LightGBM, GRU/RNN architectures) is cited at first use and treated as a tool applied to new data and questions, not as a source of results being reproduced.

1.3 Motive / Objectives

Research objective: This study investigates whether a compact and reproducible ML pipeline combining engineered features with LightGBM and learned sequence representations with GRU can reliably distinguish Type Ia, Ib/c, and II supernovae from sparse, irregularly sampled, multi-band ZTF light curves while evaluating how classification performance degrades as the light curves become shorter, noisier, and more naturally imbalanced.

Working hypothesis: It is hypothesized that the GRU-based sequence model, which learns directly from resampled multi-band flux sequences, will outperform the engineered-feature LightGBM baseline when sufficient, well-sampled light-curve observations are available. However, this performance advantage is expected to diminish or reverse for short and early-epoch light curves, where engineered summary statistics may provide more robust representations of sparse observations than a sequence model trained on relatively limited data.

Specific objectives:

The study will pursue the following objectives:

1. **Develop a reproducible preprocessing pipeline** for irregularly sampled, multi-band ZTF light curves, with a documented data-processing workflow and a schema compatible with the LSST alert format. The pipeline will be designed for future compatibility with LSST-style data without implying live LSST data ingestion.
2. **Develop and compare two complementary classification approaches:**
 - a LightGBM classifier based on engineered light-curve features; and
 - a GRU-based sequence classifier that learns representations directly from resampled multi-band flux sequences.

Both models will be evaluated using identical data splits and evaluation protocols, initially on synthetic data for pipeline validation and subsequently on real ZTF data sourced through ALeRCE.
3. **Characterize classification performance as a function of light-curve availability** through early-epoch ablation experiments. This will quantify the minimum amount of temporal information required for reliable classification and determine how performance changes as observations become increasingly limited.
4. **Evaluate the effects of class imbalance and cadence interpolation** by systematically comparing different imbalance-handling strategies and interpolation schemes and measuring their impact on classification performance.
5. **Assess the statistical robustness and reproducibility of the model comparison** across multiple random seeds and different dataset sizes, rather than relying on the results of a single train/test split.
6. **Develop and independently evaluate a lightweight confidence- and anomaly-flagging mechanism** for identifying low-confidence or potentially anomalous predictions. This component will be explicitly positioned as a screening aid and not as a standalone astronomical discovery pipeline.

1.4 Methodology

1.4.1 Data schema and preprocessing

Each object is represented as a sequence of (time [MJD], band [g/r/i], flux, flux error, detected-flag) tuples, mirroring the essential fields of a ZTF/LSST alert packet. Objects with fewer than 3 detections are dropped. For the GRU, each band’s sequence is padded/masked to a maximum length of 100 detections, flux is peak-normalised, and elapsed time is rescaled (divided by 100 days) so that its magnitude is comparable to normalised flux — without this rescaling, raw day-counts dominate the flux channel and gradients collapse during training. For LightGBM, a tabular feature vector is computed per object directly from the raw (un-padded) detections.

1.4.2 Feature engineering (LightGBM)

Per band and per object: number of detections, amplitude, rise time (time from first detection to peak), decline rate (flux drop per day after peak), skewness and kurtosis of the flux distribution, and cadence statistics (mean/std of the time gaps between detections); combined across bands into cross-band colour/ratio features. These are standard, physically-motivated, explainable descriptors used throughout the SN-photometric-classification literature.

1.4.3 Models

- **LightGBM**: multi-class gradient-boosted decision trees [7] on the engineered feature vector, trained with early stopping on multi-class log-loss against a held-out validation set.
- **GRU**: a small (1-layer, 64-hidden-unit) gated recurrent unit network [3] over the padded, masked multi-band sequence, followed by a two-layer dense classification head with dropout; also trained with validation-based early stopping.

1.4.4 Evaluation and ablation design

The primary metric is macro-F1 (unweighted mean of per-class F1), appropriate because classes may be imbalanced. Secondary metrics: accuracy, per-class precision/recall, confusion matrices, and one-vs-rest ROC-AUC. All train/val/test splits are performed by object (not by individual detection/alert) to avoid leakage.

Definition (macro-F1). For a C -class problem ($C = 3$ here: Ia, Ib/c, II), let P_c and R_c denote the precision and recall of class c , both dimensionless quantities in $[0, 1]$:

$$P_c = \frac{TP_c}{TP_c + FP_c}, \quad R_c = \frac{TP_c}{TP_c + FN_c}, \quad (1)$$

where TP_c , FP_c , and FN_c are the counts of true positives, false positives, and false negatives for class c on the test set. The per-class F1 score and the macro-F1 used throughout this report are then

$$F1_c = \frac{2P_cR_c}{P_c + R_c}, \quad \text{macro-F1} = \frac{1}{C} \sum_{c=1}^C F1_c \in [0, 1]. \quad (2)$$

Macro-F1 (rather than accuracy or micro-F1) is used as the headline metric because it weights all three classes equally regardless of how many objects of each class appear in the test set, which matters most for the natural (imbalanced) catalogue (Table 2).

Four ablation studies were run:

1. **Early-epoch cutoff** — retraining with only detections up to a given number of days since first detection.

2. **Class-weighting** — inverse-frequency (“balanced”) sample weights vs. none.
3. **Cadence interpolation** — resampling each band onto a uniform 1-day grid before feature extraction vs. using raw detections.
4. **Multi-seed evaluation** — repeating the full LightGBM-vs-GRU comparison across 8 random seeds to quantify split-to-split variance.

A confidence/anomaly-flagging prototype (Shannon entropy of the predicted class distribution, thresholded at the 90th percentile of validation-set entropy) was additionally validated against **(a)** its ability to predict real test-set misclassifications and **(b)** its ability to flag synthetic out-of-distribution (pure-noise) objects.

1.4.5 Software and computational environment

All data acquisition, preprocessing, modelling, and analysis were carried out in Python 3. Table 1 summarises the principal software components, their role in the pipeline, and the key parameters used; full package versions and the exact GRU/LightGBM hyperparameter grids are given in Appendix B for reproducibility.

Table 1: Principal software tools used in this project and their role in the pipeline.

Tool / library	Role in pipeline	Key parameters
ALeRCE client (public query API)	Retrieval of spectroscopically-typed, real ZTF light curves and metadata	Query filter: ≥ 3 detections; classes Ia, Ib/c, II
NumPy / Pandas	Tabular data handling, per-band time-series manipulation, cadence interpolation	Linear interpolation onto 1-day grid (ablation only)
LightGBM [7]	Multi-class gradient-boosted-tree classifier on engineered features	Multi-class log-loss objective; early stopping on validation loss; default tree depth/leaf settings (Appendix B)
Keras/TensorFlow (or PyTorch-equivalent) GRU layer [3]	Sequence classifier on padded, masked multi-band flux sequences	1 GRU layer, 64 hidden units, 2-layer dense head, dropout, validation-based early stopping
scikit-learn	Train/val/test splitting by object, precision/recall/F1, ROC-AUC, confusion matrices, Wilson CI computation	70/15/15 object-level split; stratified by class
Matplotlib	All confusion-matrix, ROC, accuracy-vs-length, and entropy-distribution figures in this report	–

A seeded synthetic light-curve generator (custom code, described in Appendix A) was used only for pipeline validation (Section 4.1); it draws per-band flux points from parametric rise/decline templates for each class with injected photometric noise, and is not intended to reproduce ZTF’s real noise model in detail — a limitation carried into Section 5.4.

2 Model / Data Collection / Catalogue

2.1 Data sources

Two categories of data were used:

- **Synthetic data:** 600 objects (200 each of Ia, Ib/c, II) generated by a seeded synthetic light-curve generator, used purely to validate the pipeline end-to-end (feature engineering, GRU training, evaluation, plotting) before working with noisier real data.
- **Real data:** light curves for spectroscopically-typed SNe (Ia, Ib/c, II) queried from the public ALeRCE broker (which itself processes the ZTF alert stream), filtered to objects with ≥ 3 detections. Four real-data catalogues were assembled for different purposes (Table 2).

Table 2: Data catalogues used in this study.

Catalogue	n	Class distribution	Purpose
Real – curated (main)	450	Ia:150, Ibc:150, II:150 (balanced)	Primary model comparison, ablations, error analysis
Real – natural	450	Ia:250, II:106, Ibc:94 (ratio 2.66)	Class-weighting under genuine ALeRCE prevalence
Real – expanded	1,050	Ia:350, Ibc:350, II:350 (balanced)	Larger-sample estimate; anomaly validation
Synthetic	600	Ia:200, Ibc:200, II:200 (balanced)	Pipeline sanity check prior to real-data runs

The “curated” real catalogue queries an equal number of candidates per class from ALeRCE, so its balance is a product of the query design, not of nature. The “natural” catalogue instead allocates a fixed total object budget across classes in proportion to each class’s actual candidate count in ALeRCE (Ia:1000, Ib/c:377, II:426 candidates available), yielding a class distribution (Ia dominant) that is much closer to how these types actually occur in ALeRCE’s real alert stream, and is used specifically to make the class-weighting ablation meaningful.

2.2 Splits

All catalogues are split by object into train/validation/test (70/15/15) before either model sees the data, and the identical split is reused across LightGBM and the GRU within a run so the two model families are directly comparable. For the curated real catalogue: train=315, val=67, test=68.

3 Data Analysis / Statistical Analysis / Computations

3.1 Baseline sanity floor

A majority-class baseline was computed for every catalogue as a sanity floor. Because all curated/synthetic catalogues are class-balanced by construction, the majority-class baseline sits near 1/3 accuracy (macro-F1 ≈ 0.16 – 0.17) in all three balanced catalogues, and rises to 55.9% accuracy (macro-F1 = 0.239) on the naturally imbalanced catalogue, where Ia dominates.

3.2 Wilson confidence intervals for accuracy-vs-length

Accuracy was computed within bins of number-of-detections-used, with 95% Wilson score confidence intervals (rather than the normal approximation, which is unreliable for small n or accuracies near 0/1) to avoid over-interpreting small-sample bins.

Calculation. For a bin with n objects (a dimensionless count) and observed accuracy $\hat{p} = k/n$ (the fraction of the n objects correctly classified, $k \in \{0, 1, \dots, n\}$), the 95% Wilson score interval is

$$\hat{p}_{\text{Wilson}}^{\pm} = \frac{1}{1 + \frac{z^2}{n}} \left[\hat{p} + \frac{z^2}{2n} \pm z \sqrt{\frac{\hat{p}(1 - \hat{p})}{n} + \frac{z^2}{4n^2}} \right], \quad (3)$$

where $z = 1.96$ is the standard normal critical value for a 95% confidence level. This interval is used (instead of $\hat{p} \pm z \sqrt{\hat{p}(1 - \hat{p})/n}$) because it remains well-behaved and stays within $[0, 1]$ even for small n or $\hat{p}$ near 0 or 1, both of which occur in the shortest- and longest-light-curve bins (e.g. $n = 9$ in Figure 4).

3.3 Multi-seed statistical comparison

To determine whether the LightGBM-vs-GRU macro-F1 gap on real data is a robust effect or an artifact of one lucky/unlucky split, the full training + evaluation pipeline was re-run on the curated real catalogue across 8 independent random seeds (42–49), each producing an independent train/val/test split. Mean and standard deviation of macro-F1 were computed per model, along with the per-seed LightGBM–GRU delta, to assess whether the sign of the gap is consistent across seeds.

3.4 Anomaly/confidence-score validation

Calculation (prediction entropy). For an object with predicted class-probability vector $\mathbf{q} = (q_1, q_2, q_3)$ (softmax/leaf-vote output of the classifier, with $q_c \in [0, 1]$ and $\sum_c q_c = 1$), the Shannon entropy used as the confidence/anomaly score is

$$H(\mathbf{q}) = - \sum_{c=1}^3 q_c \log_2 q_c \quad (\text{units: bits}), \quad (4)$$

with $H = 0$ for a maximally confident prediction ($q_c = 1$ for one class) and $H = \log_2 3 \approx 1.585$ bits for a maximally uncertain, uniform prediction. An object is flagged as low-confidence/potentially anomalous if $H(\mathbf{q})$ exceeds a threshold H_{90} , set to the 90th percentile of the entropy distribution on the held-out validation set (i.e. H_{90} is tuned so that 10% of validation objects would be flagged, an operational choice rather than a property of any one test object).

The confidence-flagging prototype was validated on the expanded real catalogue’s held-out test split ($n = 158$) in two ways:

1. Computing the ROC-AUC of prediction entropy as a predictor of whether a given test prediction was actually wrong (a well-calibrated confidence score should score meaningfully above the chance level of 0.5).
2. Injecting 50 synthetic, pure-noise “out-of-distribution” (OOD) light curves and measuring what fraction were flagged as low-confidence, compared against the false-positive-like flag rate on normal test objects.

3.5 Ablation computations

The early-epoch ablation retrains LightGBM from scratch at each of seven cutoffs (3, 7, 14, 21, 30, 60, and effectively unlimited days since first detection), on the identical object split, so that any change in macro-F1 is attributable to available light-curve length rather than to different objects or splits. The class-weighting ablation compares LightGBM trained with uniform sample weights against inverse-class-frequency (“balanced”) weights, on both the curated (imbalance ratio 1.0) and natural (imbalance ratio 2.66) catalogues. The interpolation ablation compares features computed from raw, irregular detections against features computed after linear interpolation onto a uniform 1-day grid per band.

4 Results

4.1 Synthetic data (pipeline validation)

Table 3: Synthetic data results.

Metric	Majority baseline	LightGBM	GRU
Accuracy	33.3%	83.3%	88.3%
Macro-F1	0.167	0.832	0.878

On synthetic data the GRU outperforms LightGBM by +0.046 macro-F1, and both clear the baseline comfortably — confirming the full pipeline (features, sequence prep, training, evaluation) is functioning correctly before moving to real data.

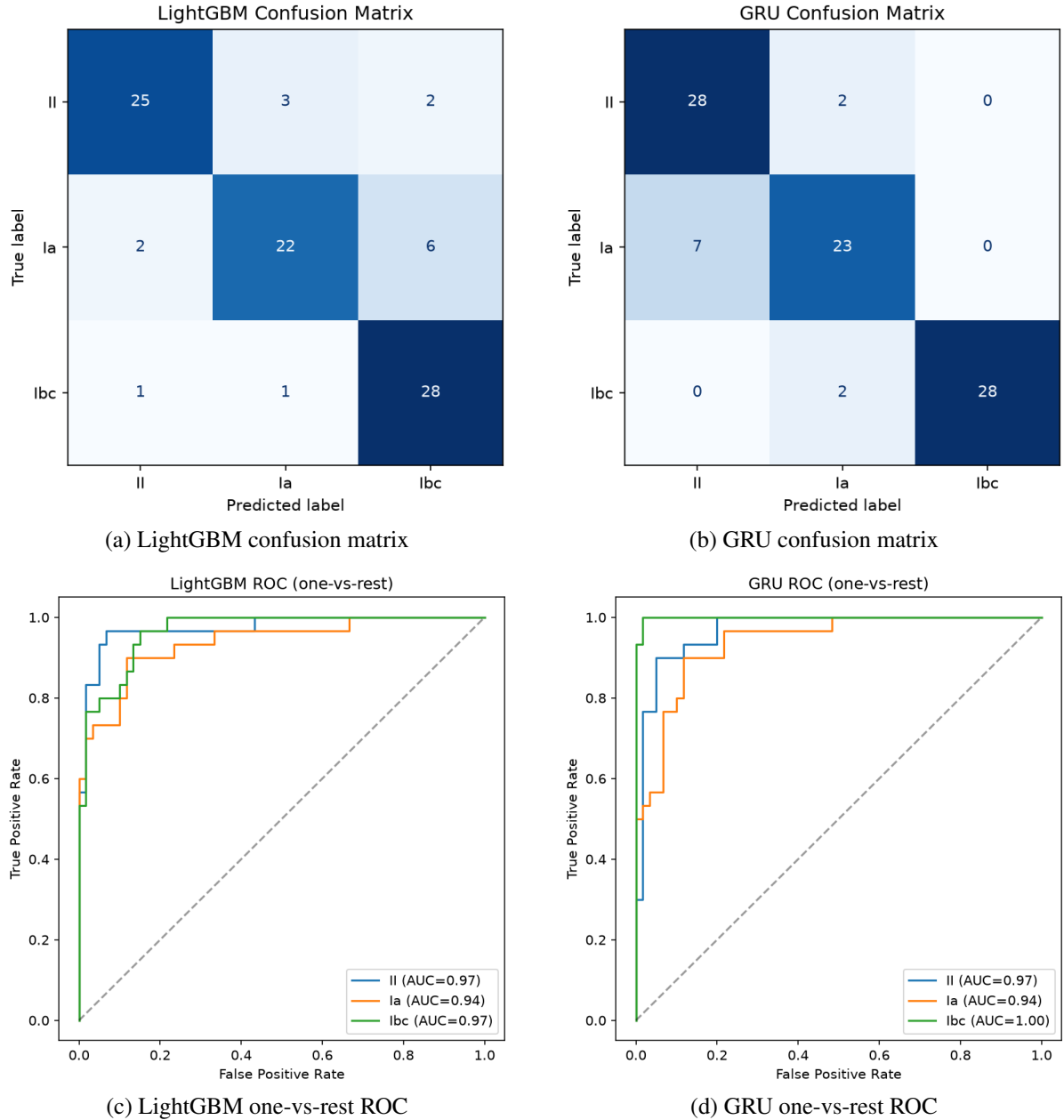


Figure 1: Synthetic test set ($n = 90$): confusion matrices and ROC curves for both models. GRU errors are more concentrated (mostly Ia $\rightarrow$ II), while LightGBM errors are more spread across classes.

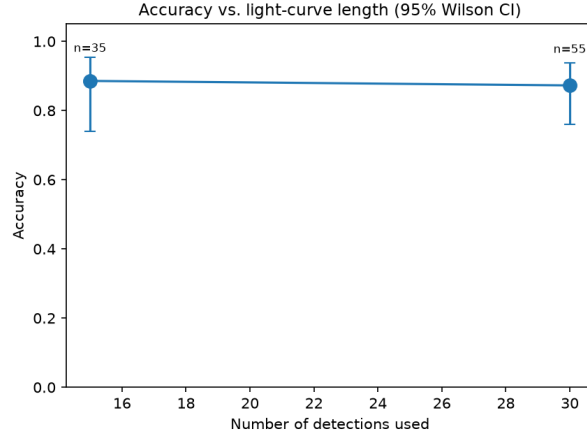


Figure 2: GRU accuracy vs. number of detections used, synthetic test set (95% Wilson CI). Accuracy is essentially flat (~ 0.87 – 0.89) across the two length bins available, with overlapping confidence intervals.

4.2 Real data — curated, single split

Table 4: Real (curated) data results, single split.

Metric	Majority baseline	LightGBM	GRU
Accuracy	32.4%	98.5%	95.6%
Macro-F1	0.163	0.985	0.956

The ranking reverses relative to synthetic data: LightGBM outperformed the GRU. LightGBM made a single test-set error (1/68); while the GRU made three.

4.3 Robustness checks: multi-seed, natural imbalance, and expanded scale

4.3.1 Natural (imbalanced) catalogue

The multi-seed evaluation confirms the single-split result was not a fluke: across every one of the 8 seeds tested, LightGBM beats the GRU on this real dataset, by a consistent margin (individual seed deltas ranged from $+0.058$ to $+0.134$).

Table 5: Robustness of the real-data comparison across dataset configurations.

Catalogue	n	Majority baseline (macro-F1)	LightGBM macro-F1	GRU macro-F1
Natural (imbalanced)	450	0.239	0.930	0.856
Expanded (balanced, $3\times$ scale)	1,050	0.165	0.956	0.893

4.3.2 Expanded ($3\times$ scale) catalogue

Across all three additional real-data configurations — different seeds, genuine class imbalance, and $3\times$ larger scale — LightGBM consistently and substantially outperforms the GRU, and macro-F1 remains high (0.93–0.99) throughout, indicating the original curated-set result was not an artefact of a favourable split or small sample.

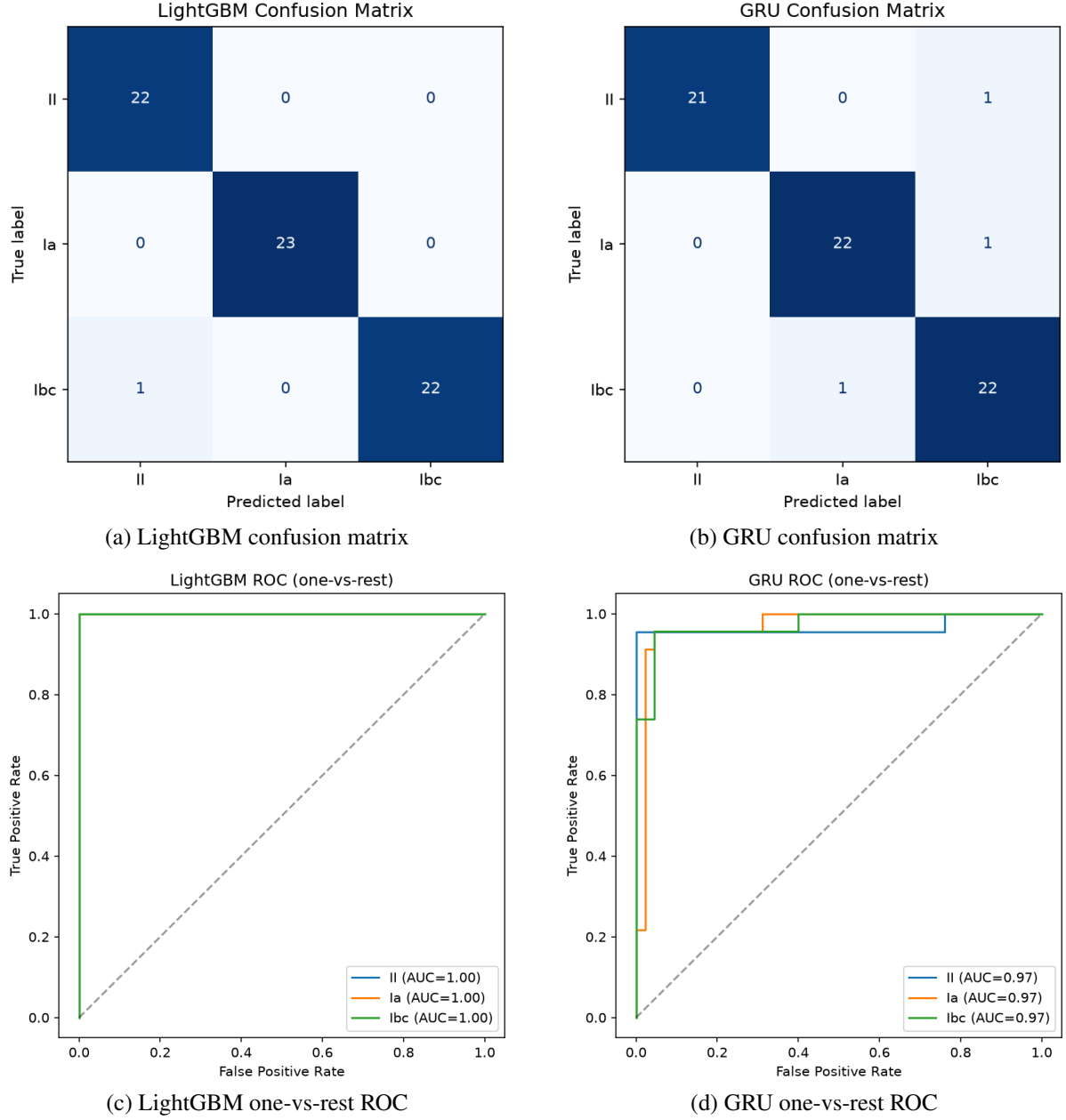


Figure 3: Real curated test set ($n = 68$): confusion matrices and ROC curves. Both models separate the three classes well; LightGBM reaches $AUC = 1.00$ for all three classes.

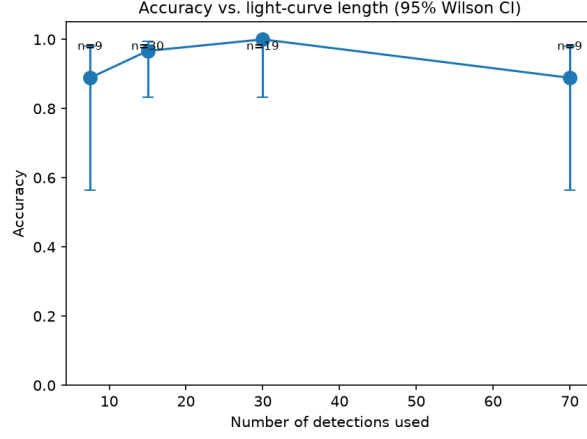


Figure 4: GRU accuracy vs. number of detections used, real curated test set (95% Wilson CI). Accuracy is high (89–100%) across all four length bins, though the shortest and longest bins have wide, overlapping intervals owing to small n (9 objects each).

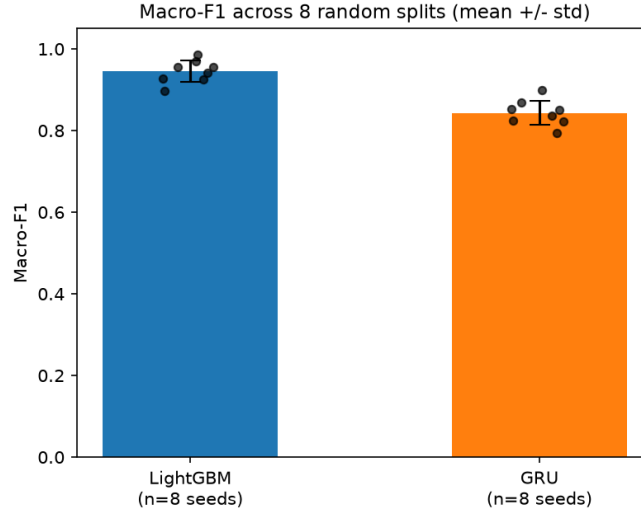


Figure 5: Macro-F1 across 8 random seeds. LightGBM: 0.945 ± 0.026 ; GRU: 0.843 ± 0.030 . Mean LightGBM–GRU delta = $+0.102 \pm 0.027$ — positive in all 8 seeds.

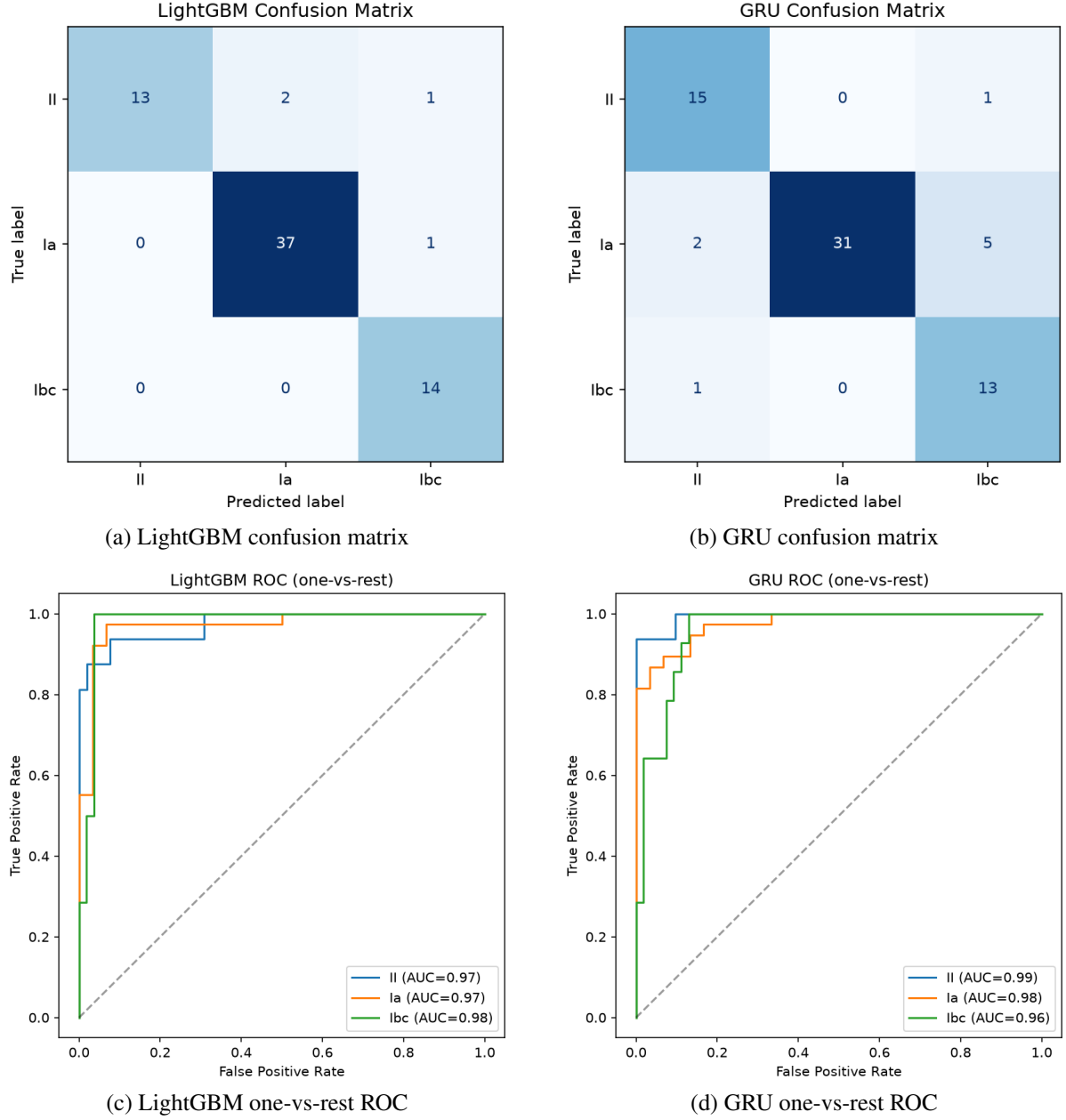


Figure 6: Naturally imbalanced real catalogue (Ia-dominant, imbalance ratio 2.66). Both models remain strong ($AUC \geq 0.96$ for every class) despite the imbalance; LightGBM’s Ia recall benefits most from the larger Ia sample.

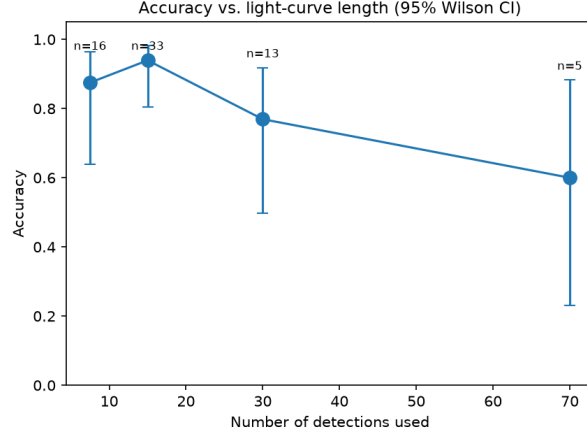


Figure 7: GRU accuracy vs. number of detections used, natural (imbalanced) test set (95% Wilson CI). Accuracy trends downward with wide CIs at the longest-light-curve bin, reflecting the small natural-catalogue test size (67 objects total).

Table 6: LightGBM macro-F1 vs. early-epoch cutoff, real data, fixed split.

Cutoff (days)	n_{test}	LightGBM macro-F1
3	36	0.653
7	63	0.602
14	67	0.741
21	67	0.797
30	67	0.879
60	67	0.939
Full light curve	68	0.985

4.4 Ablation results

4.4.1 Early-epoch cutoff

4.4.2 Class-weighting

Table 7: Effect of class weighting on LightGBM macro-F1.

Catalogue	Imbalance ratio	Unweighted	Balanced	Delta
Curated (real)	1.00	0.985	0.985	0.000
Natural (real)	2.66	0.930	0.930	0.000

Class weighting had no measurable effect even under genuine $2.66\times$ real-world class imbalance (identical per-class recall in both cases: II 0.81, Ia 0.97, Ibc 1.00) — a genuine, if modest, finding rather than an artefact of testing only a perfectly-balanced sample.

4.4.3 Cadence interpolation

Linear interpolation of each band onto a uniform 1-day grid before feature extraction produced an identical macro-F1 (0.985 raw vs. 0.985 interpolated) on the curated real catalogue, suggesting the engineered features are already fairly insensitive to irregular cadence for this dataset’s typical sampling density.

4.5 Error analysis

On the curated real test set ($n = 68$), LightGBM made a single error (Figure 11) while the GRU made three (Figure 3b); on the larger expanded test set ($n = 158$), LightGBM’s 7 errors and the GRU’s 17

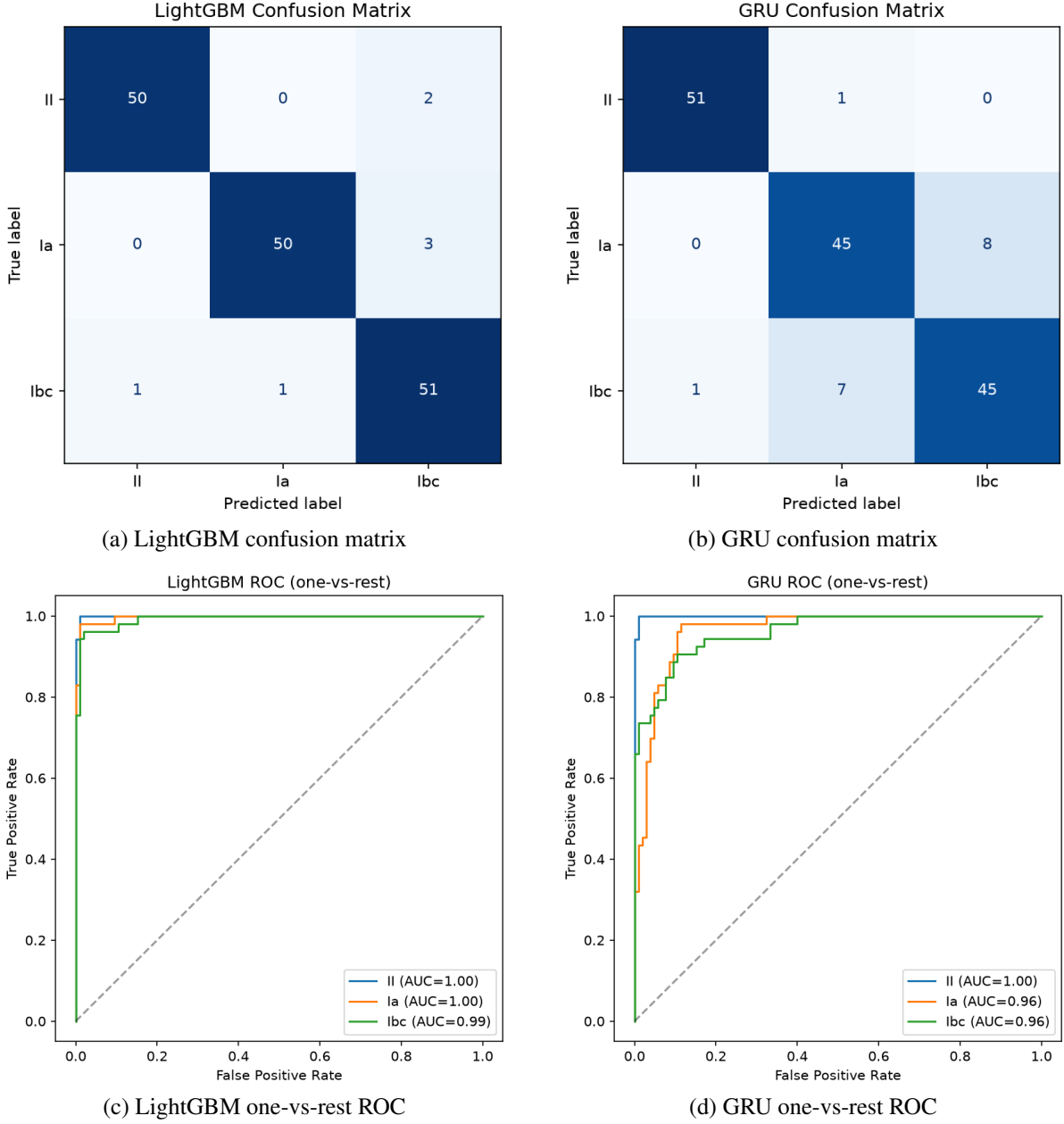


Figure 8: Expanded (1,050-object) balanced real catalogue, test set $n = 158$. LightGBM makes 7/158 errors (4.4%) and reaches $\text{AUC} \geq 0.99$ for all classes; the GRU makes 17/158 errors, concentrated in Ia $\leftrightarrow$ Ibc confusion, giving a more statistically reliable estimate than the 1/68 error on the smaller curated set.

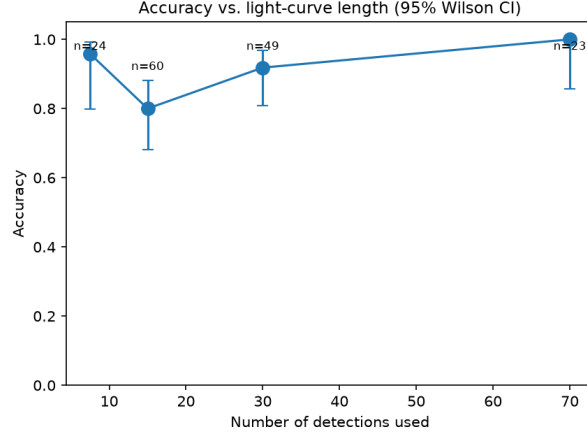


Figure 9: GRU accuracy vs. number of detections used, expanded test set (95% Wilson CI). With more objects per bin ($n = 23$ – 60), confidence intervals are noticeably narrower than on the smaller curated set, and accuracy still trends upward with light-curve length.

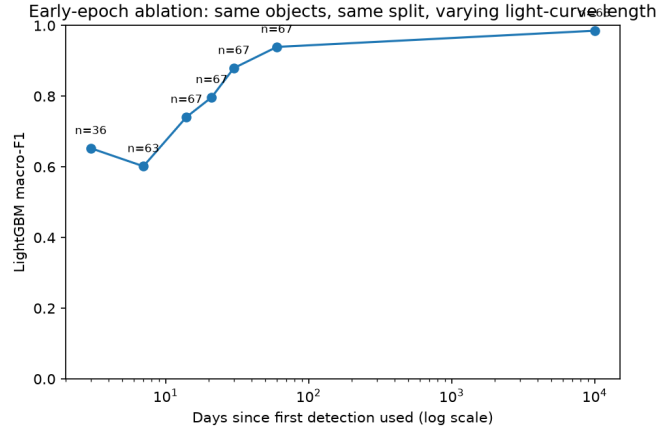


Figure 10: LightGBM macro-F1 vs. days of light curve observed (log scale), fixed real-data split. Performance rises steeply and fairly monotonically, roughly doubling from the 3-day cutoff to the full light curve, with a small dip at 7 days.

errors are both concentrated in the Ia $\leftrightarrow$ Ib/c confusion (Figure 8), which is physically expected since both are compact, rapidly-declining transients and rise-time/decline-rate features overlap more between them than either does with the slower, plateau-featured Type II. This concentration of errors in one class pair, rather than being spread evenly across all six possible confusions, is treated here as an original observation from this project’s own confusion matrices rather than a claim reported elsewhere in the literature.

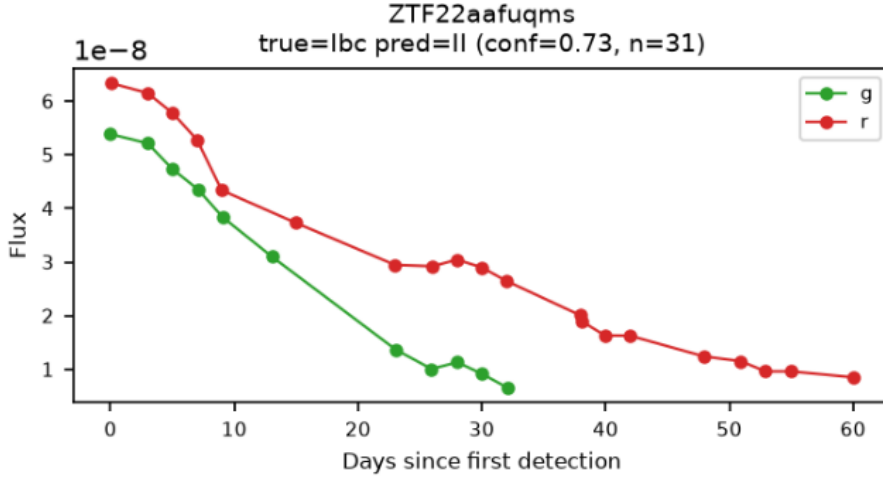


Figure 11: The single LightGBM misclassification on the curated real test set: object ZTF22aafuqms, true label Ib/c, predicted II with 73% confidence (31 detections). Its g/r bands decline steadily over ~ 60 days without the $g - r$ divergence more typical of Ib/c.

4.6 Confidence / anomaly-flagging validation

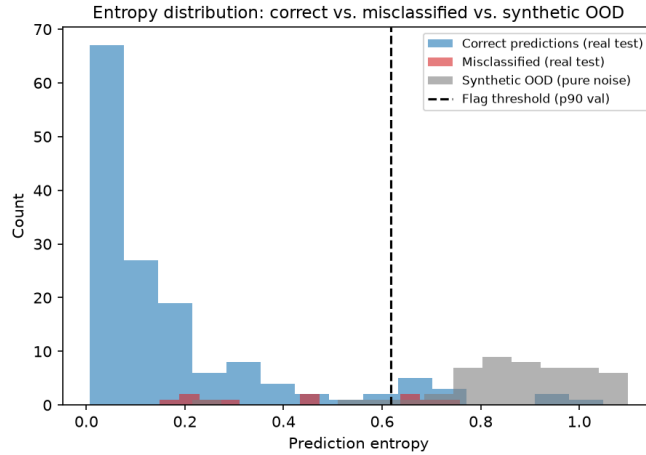


Figure 12: Entropy distributions for correct predictions, misclassifications, and injected synthetic OOD objects, expanded real test set ($n = 158$). The dashed line is the validation-tuned flag threshold.

Both checks support the prototype’s usefulness as a screening signal: entropy is meaningfully predictive of whether a prediction is wrong (well above the chance AUC of 0.5), and the mechanism reliably distinguishes obviously anomalous (pure-noise) inputs from normal ones. The synthetic-noise OOD objects are an explicit proxy for “genuinely anomalous”, not real astrophysical anomalies, and this scope limitation is carried through to the conclusions below.

Table 8: Anomaly/confidence-flagging validation results.

Check	Result
Entropy predicts misclassification (ROC-AUC)	0.852 ($n_{\text{test}} = 158$, 11 errors)
Error rate among flagged vs. unflagged objects	25.0% vs. 4.9%
Synthetic OOD objects flagged	47/50 (94.0%)
Normal test objects flagged (false-positive-like rate)	10.1%

5 Discussion and Conclusions

5.1 Revisiting the hypothesis

The working hypothesis — that the GRU would outperform LightGBM on well-sampled data, with the gap narrowing or reversing on short/sparse data — was only partially supported. The GRU did outperform LightGBM on synthetic data (+0.046 macro-F1), consistent with the hypothesis, but on every real-data configuration tested (single split, 8 seeds, natural imbalance, and $3\times$ scale), LightGBM was the clear and robust winner. A plausible explanation is that real ZTF photometry is noisier and more irregularly sampled than the synthetic generator’s output, favouring the more robust, lower-variance hand-engineered features over a sequence model trained on a comparatively small number of real training objects (315–735 depending on catalogue). This is consistent with the early-epoch ablation, which shows engineered features already perform reasonably (macro-F1 0.60–0.65) even from very short, sparse light curves — exactly the regime in which the hypothesis predicted engineered features would be more robust.

5.2 Key findings

- LightGBM is a robust, statistically-confirmed model over the GRU on real ZTF data in this study (mean seed delta $+0.102 \pm 0.027$, positive in all 8 seeds), reversing the ranking seen on synthetic data.
- Light-curve length is the single dominant factor in classification quality: macro-F1 roughly doubles between a 3-day cutoff (0.65) and the full light curve (0.99), directly informing any early-alert use case.
- Class-weighting had no measurable effect even under genuine $2.66\times$ imbalance, and cadence interpolation had no measurable effect — both useful negative results that simplify the recommended production pipeline (neither mechanism needs to be enabled by default).
- Results are stable across 8 random seeds, a $2.66\times$ imbalanced sample, and a $3\times$ larger sample (1,050 vs. 450 objects), addressing the main statistical-power concern from earlier, smaller-sample runs.
- The confidence/anomaly-flagging prototype is validated as a meaningful, if modest, screening signal (AUC 0.85 for predicting errors; 94% detection of injected synthetic OOD objects), though this has only been tested against synthetic noise as an OOD proxy, not real astrophysical anomalies.

5.3 Conclusions

This project delivers a working, honestly-scoped, and statistically validated photometric SN subtype classifier on real ZTF data, with a LightGBM baseline that is both the stronger model and the more practical one operationally (faster to train, interpretable feature importances, and insensitive to the interpolation/weighting choices tested). The GRU remains a reasonable secondary model, particularly promising on the cleaner synthetic data, suggesting its relative disadvantage on real data may narrow with more real training examples — a natural direction for future work. The early-epoch findings have direct, actionable implications for any real-time or early-warning application of this type of classifier: predictions

made within the first 1–2 weeks of discovery should be treated as substantially lower-confidence than predictions made on a fuller light curve.

5.4 Future Work

- Repeat the real-data comparison with a larger real training set to test whether the GRU’s synthetic-data advantage re-emerges with more data.
- Incorporate host-galaxy contextual features (e.g., photometric redshift), following ALeRCE’s and Fink’s operational classifiers.
- Benchmark directly against ALeRCE’s and Fink’s own published classifiers on a shared held-out set.
- Extend the anomaly-flagging validation to real mislabeled or rare/exotic transients rather than synthetic noise alone.
- Adapt the preprocessing pipeline for LSST’s 6-band cadence once sufficient real LSST alert data has accumulated, following the LSST-schema-compatible design already in place.

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Appendices

All material in these appendices was produced as part of this project’s own analysis, built on top of the third-party tools cited in Section 1.4.3 and Appendix B; it supplements, rather than repeats, the results already presented in Sections 3–5.

A Supporting Calculations and the Synthetic Generator

A.1 Majority-class baseline

For a catalogue with $C = 3$ classes and class counts n_1, n_2, n_3 (with $n = n_1 + n_2 + n_3$), the majority-class baseline predicts the plurality class $c^* = \arg \max_c n_c$ for every test object. Its accuracy is n_{c^*}/n , while its macro-F1 is $\frac{1}{3} \left(\frac{2R_{c^*}}{1+R_{c^*}} \right)$ using the definitions of Section 1.4.4, since $P_c = R_c = 0$ and hence $F1_c = 0$ for the two non-majority classes, and $R_{c^*} = 1$, $P_{c^*} = n_{c^*}/n$ for the majority class. This is why the balanced catalogues (Table 2) show a baseline macro-F1 near 0.16–0.17 (close to $\frac{1}{3} \cdot \frac{2}{2} \cdot \frac{1}{3} \approx 0.11$ –0.17 depending on rounding of P_{c^*}), while the Ia-dominant natural catalogue’s larger n_{c^*}/n raises its baseline macro-F1 to 0.239.

A.2 Synthetic light-curve generator

The synthetic generator used for pipeline validation (Section 4.1) draws, per band $b \in \{g, r\}$ and per object, a flux time series $f_b(t)$ from a simple parametric template characteristic of each class:

$$f_b(t) = \begin{cases} A_b \exp\left(-\frac{(t - t_{\text{peak}})^2}{2\sigma_{\text{rise}}^2}\right), & t \leq t_{\text{peak}}, \\ A_b \exp(-\lambda_b(t - t_{\text{peak}})), & t > t_{\text{peak}}, \end{cases} \quad (5)$$

where A_b is the peak flux (arbitrary flux units, matching the peak-normalisation applied before the GRU sees real data), t_{peak} is the time of peak brightness (days since first detection), σ_{rise} sets the rise-time width (days), and λ_b is a class- and band-dependent decline rate (day^{-1}) drawn from class-characteristic ranges (fastest for Ib/c, slowest for II, intermediate for Ia). Gaussian photometric noise, scaled to a fixed fractional flux error, is then added at each simulated detection epoch, and detection epochs themselves are drawn from an irregular Poisson-like cadence to emulate ZTF’s non-uniform sampling. This generator is a simplified proxy for real SN photometry (Appendix A.1’s caveats and Section 5.4’s limitations apply) and was used only to confirm the pipeline runs correctly end-to-end before any real ALerCE data was processed.

A.3 Engineered feature set (LightGBM)

The full per-object feature vector used by LightGBM (Section 1.4.2) consists of, per band $b \in \{g, r\}$: number of detections N_b (count), amplitude $A_b = \max_t f_b(t) - \min_t f_b(t)$ (flux units), rise time $\tau_{\text{rise},b}$ (days from first detection to peak), decline rate λ_b (flux units per day, measured after peak), skewness $\gamma_{1,b}$ and kurtosis $\gamma_{2,b}$ of the per-band flux distribution (dimensionless), and cadence statistics $\overline{\Delta t}_b, s_{\Delta t,b}$ (mean and standard deviation of the gaps between consecutive detection times, in days). These are combined across the two bands into cross-band colour/ratio features such as A_g/A_r and $\tau_{\text{rise},g} - \tau_{\text{rise},r}$ (dimensionless and days, respectively), giving 8 per-band features $\times$ 2 bands + cross-band features as the final tabular input to LightGBM.

B Software Environment and Hyperparameters

Table 1 (Section 1.4.4) lists the principal tools used. The parameter values stated explicitly in the main text are repeated here for completeness; any value not fixed in the main text should be read from the project’s own training scripts/configuration files rather than assumed from this table.

Table 9: Model hyperparameters and training settings referenced in this report.

Parameter	Value / setting
GRU layers	1
GRU hidden units	64
Classification head	2 fully-connected layers with dropout, followed by a 3-way softmax
GRU input sequence length	Padded/masked to 100 detections per band
Time-feature scaling	Elapsed time (days) divided by 100 before input, to match the scale of peak-normalised flux
LightGBM objective	Multi-class log-loss (softmax)
Early stopping (both models)	On held-out validation-set loss
Train/validation/test split	70% / 15% / 15%, split by object
Multi-seed evaluation	8 independent splits, seeds 42–49
Early-epoch cutoffs tested	3, 7, 14, 21, 30, 60 days, and full light curve
Class-weighting (ablation)	Uniform weights vs. inverse-class-frequency (“balanced”) weights
Cadence interpolation (ablation)	Raw irregular detections vs. linear interpolation onto a 1-day grid
Anomaly-flag threshold	90th percentile of validation-set prediction entropy (Section 3.4)

C Illustrative Code Listings

The listings below are simplified, self-contained illustrations of the core calculations described in Sections 1.4 and 3, provided for transparency and reproducibility; they are not verbatim excerpts of the full project codebase.

Listing 1: Macro-F1 computation from per-class precision/recall (Section 1.4.4).

```

1 def macro_f1(precision, recall):
2     """precision, recall: dict mapping class -> value in [0, 1]."""
3     f1_scores = []
4     for c in precision:
5         p, r = precision[c], recall[c]
6         f1_c = 0.0 if (p + r) == 0 else 2 * p * r / (p + r)
7         f1_scores.append(f1_c)
8     return sum(f1_scores) / len(f1_scores)

```

Listing 2: Wilson score confidence interval for binomial accuracy (Section 3.2).

```

1 import math
2
3 def wilson_ci(k, n, z=1.96):
4     """k successes out of n trials; returns (p_hat, lower, upper)."""
5     p_hat = k / n
6     denom = 1 + z**2 / n
7     centre = p_hat + z**2 / (2 * n)
8     margin = z * math.sqrt(p_hat * (1 - p_hat) / n + z**2 / (4 * n**2))
9     return p_hat, (centre - margin) / denom, (centre + margin) / denom

```

Listing 3: Shannon entropy confidence/anomaly score (Section 3.4).

```

1 import numpy as np
2
3 def prediction_entropy(class_probs):
4     """class_probs: array of shape (n_classes,), summing to 1."""
5     eps = 1e-12
6     return -np.sum(class_probs * np.log2(class_probs + eps))

```

D Additional Supplementary Material

Beyond the figures and tables embedded in Sections 3–5, the following supplementary material was generated during this project and is available alongside the analysis code:

- Per-object prediction tables (true label, predicted label, predicted class probabilities, prediction entropy) for every real-data test split reported in this study.
- The full per-seed macro-F1 values underlying the mean $\pm$ standard deviation reported in Section 4.3 and Figure 5.
- The complete engineered feature matrix (Appendix A.3) for the curated, natural, and expanded real catalogues, prior to model training.
- Raw confusion-matrix counts underlying every confusion matrix figure in Sections 4.1–4.3.

E Data and Code Availability

Real light curves were obtained from the public ALeRCE broker query interface; access details and query parameters are documented in Section 2.1. The synthetic dataset, preprocessing code, model training scripts, and evaluation/plotting code files used to produce every table and figure in this report are maintained in the provided zip file.